



Cognitive enhancement through competency-based programming education: a 12-year longitudinal study

Dunhong Yao^{1,2,3} · Jing Lin^{1,2,3}

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Abstract

Programming education consistently faces challenges in bridging theory with practice and fostering students' cognitive competencies. This 12-year longitudinal study (2011–2023) investigates an innovative competency-based teaching model in university C programming education that integrates six educational theories into a coherent framework with three dimensions (theoretical, practical, innovative), four integration mechanisms, and five combinatorial strategies. Using a mixed-methods approach with a quasi-experimental design, we studied 4,051 undergraduate students from a Chinese university. Results revealed significant enhancement in students' cognitive abilities, as measured by Raven's Standard Progressive Matrices ($t(350)=8.76$, $p<0.001$, $d=0.68$), which strongly correlated with improved academic performance ($r=0.62$), computational thinking ($r=0.71$), and problem-solving skills ($r=0.67$). The model creates multiple pathways for cognitive development through synergistic interactions between components, promoting collaboration and self-directed learning with effects extending beyond graduation. Multiple regression analysis identified three key predictors of cognitive enhancement: classroom engagement ($\beta=0.35$), project completion ($\beta=0.28$), and participation in innovation activities ($\beta=0.22$). This study provides robust empirical evidence for the long-term efficacy of a competency-based model in programming education, presenting a transformative approach to STEM education reform particularly relevant in rapidly evolving technological landscapes.

Keywords Competency-based education · Integrated pedagogical framework · Computer programming pedagogy · Cognitive development · Mixed research methods · Longitudinal study

1 Introduction

Computer programming education serves as a cornerstone in the development of information technology professionals. Despite its importance, traditional pedagogical approaches encounter substantial challenges in effectively fostering students' cognitive capabilities and practical proficiencies (Mulopo & Fowler, 1987; Terenzini et al., 2001; Luxton-Reilly et al., 2018).

First, the disconnection between theoretical knowledge and practical application significantly impedes programming education. Traditional pedagogical approaches consistently demonstrate inadequacies in effectively integrating abstract concepts with concrete implementation skills. Consequently, students often exhibit proficiency in theoretical examinations while simultaneously struggling with authentic programming scenarios that require applied problem-solving capabilities (Vihavainen et al., 2014; Medeiros et al., 2019).

Second, outdated curriculum content fails to keep pace with technological evolution. While foundational programming languages remain essential, traditional curricula often fail to adequately incorporate contemporary programming practices or emerging industrial trends (Dung, 2024). This misalignment produces graduates who are ill-prepared for the current technological landscape, highlighting the need for innovative pedagogical strategies to bridge this gap (Puri, 2024). In the face of artificial intelligence (AI) and digital transformation, traditional programming education models have revealed severe inadequacies. According to recent industry surveys (World Economic Forum, 2023), over 60% of companies report that graduates' practical abilities fail to meet job requirements, while 90% of programming course content has an update cycle exceeding three years, significantly lagging technological advancements. These challenges are further compounded by the increasing complexity of software development and rapid evolution of programming paradigms (Gartner, 2023).

Third, insufficient practical skills development remains a major challenge. The dearth of practical opportunities and real-world project experience in many educational programs engenders graduates whose programming skills fall short of the industry standards. Students often complete their studies with theoretical knowledge but lack proficiency in debugging complex systems or collaborating on large-scale projects, thus diminishing their effectiveness in professional environments (Radermacher & Walia, 2013).

Fourth, inadequate cultivation of innovative thinking constrains students' potential. Traditional educational models often emphasize knowledge transmission at the expense of developing students' innovative capacities and problem-solving abilities. This approach limits students' ability to address novel challenges and adapt to rapidly evolving technological contexts (Alammary, 2019).

Fifth, low student engagement and motivation undermine learning effectiveness. Many students view traditional programming courses as abstract and disconnected from real-world applications, resulting in diminished motivation and engagement. This lack of enthusiasm can negatively impact learning outcomes and long-term interest in the discipline (Souza & Bittencourt, 2019; Elteгани & Butgereit, 2015).

Recent studies (2023–2024) have highlighted several promising directions to address these challenges. Technology-enhanced learning solutions such as adaptive learning systems and intelligent tutoring platforms have shown potential for addressing student engagement and personalization challenges (Gligorea et al., 2023; Zhang, 2024). Practice-industry integration has increasingly addressed the gap between theory and practice through industry-aligned curricula and real-world projects (Zhou et al., 2023; Navarro-Cota et al., 2024). Cognitive development approaches have emphasized the role of cognitive enhancement in programming education, showing strong correlations between cognitive development and programming proficiency (Berssanette & De Francisco, 2022; Bubnic et al., 2024). Collaborative learning frameworks increasingly focus on developing technical and soft skills in structured collaborative learning environments (Cunha et al., 2024; Etemi et al., 2024).

In response to these persistent challenges, Competency-Based Education (CBE) models have emerged as prominent approaches to computer science education (Malhotra et al., 2023; Clear et al., 2020). CBE models emphasize the cultivation of specific skills and knowledge that are directly applicable to practical programming tasks, thereby addressing the increasing demand for industry readiness and practicality (Palmer-Brown et al., 2017).

CBE offers multiple substantive advantages in addressing the persistent limitations of traditional programming education. First, it prioritizes demonstrable competencies over theoretical knowledge acquisition, thereby systematically narrowing the theory-practice divide. Second, its flexible modular framework facilitates rapid curricular adaptation to maintain alignment with evolving industrial requirements. Third, CBE emphasizes experiential learning through structured hands-on projects and authentic applications, directly fostering practical skill development. Fourth, its personalized learning pathways significantly enhance student engagement and intrinsic motivation through individualized progression opportunities.

Although CBE demonstrates theoretical potential, its long-term efficacy in programming education remains largely unexplored. Previous studies have attempted to improve programming education using various approaches, but most have been limited to validating the short-term effects of a single teaching method. Thus, there is an urgent need to establish a systematic teaching model and to verify its effectiveness in enhancing students' cognitive abilities and professional competencies through longitudinal research. This gap is particularly significant, given the increasing demand for adaptable IT professionals in a rapidly evolving technological landscape.

To address these critical challenges in programming education, we formulated three core research questions that guided our systematic investigation of the competency-based teaching model's effectiveness:

First, to what extent does the competency-based teaching model influence students' learning outcomes, specifically regarding academic performance, knowledge acquisition, and cognitive development?

Second, what specific impact does this model have on the development of students' cognitive abilities, particularly computational thinking (CT) and problem-solving skills?

Third, how does the model affect the development of students' self-directed learning (SDL) abilities, teamwork skills, and ethical cognition?

Based on these research questions, we formulate the following hypotheses:

H1 Compared to traditional pedagogical methods, the competency-based teaching model will significantly improve students' learning outcomes, including academic performance and knowledge acquisition.

H2 Students exposed to a competency-based model will exhibit enhanced cognitive abilities, particularly CT and problem-solving skills.

H3 The competency-based model positively affects students' SDL abilities, teamwork skills, and ethical cognition, thereby contributing to the development of well-rounded IT professionals.

By addressing these questions through a comprehensive long-term study, this research aims to provide empirical support for CBE models and offer valuable insights into programming curriculum reform and STEM education. These findings seek to provide critical guidance for talent development policies aimed at enhancing cognitive skills and present new perspectives on cultivating IT professionals adaptable to the ever-evolving demands of the industry.

This study makes several significant contributions to research and practice in programming education:

As a theoretical contribution, we present the first systematic integration of the six educational theories into a coherent competency-based teaching model for programming education. This 12-year longitudinal study provides robust empirical evidence for the model's effectiveness in enhancing students' cognitive abilities and programming competencies.

Second, as a methodological innovation, we developed a comprehensive assessment framework that combines traditional academic metrics with cognitive development measures to provide a more holistic evaluation approach for programming education outcomes.

Third, regarding practical implications, our study offers concrete guidelines for implementing competency-based programming education, including detailed teaching strategies, assessment methods, and adaptation mechanisms in different educational contexts.

The remainder of this paper is organized as follows. Section 2 presents a comprehensive review of relevant literature on competency-based programming education, cognitive development, CT, and SDL. Section 3 details the research methodology, including the development and implementation of the teaching model, the data collection methods, and the analytical approaches. Section 4 presents the results of our 12-year longitudinal study focusing on academic performance, cognitive development, and various competency indicators. Finally, Sect. 5 discusses the implications of our findings, acknowledges their limitations, and suggests directions for future research.

2 Literature review

This review critically analyzes the recent advancements in competency-based programming education, emphasizing their effects on students' cognitive abilities, CT, and SDL. These dimensions form the foundation of our proposed integrated teaching model, which aims to address the challenges prevalent in the current programming education paradigms.

2.1 Competency-based programming education and cognitive development: an integrated framework

Computer programming education consistently faces challenges in effectively nurturing students' cognitive capabilities and practical proficiencies. CBE models have emerged as prominent approaches in computer science education (Malhotra et al., 2023; Clear et al., 2020; Frezza et al., 2018). These models emphasize the cultivation of specific skills and knowledge that are directly applicable to practical programming tasks, thereby addressing the increasing demand for industry readiness and practicality (Palmer-Brown et al., 2017; Brower et al., 2017).

CBE offers several potential advantages for addressing the limitations of traditional programming education. It prioritizes demonstrable competencies over mere theoretical knowledge, potentially narrowing the theory-practice divide (Curry & Docherty, 2017; Delane & Haynes, 2021). The flexible modular framework enables rapid curricular updates to maintain alignment with industrial trends (Sidhu et al., 2021; Zhou et al., 2023). CBE emphasizes experiential learning through hands-on projects and real-world applications (Assyne et al., 2022; Saad & Zainudin, 2022). Additionally, personalized learning pathways enhance student engagement and motivation (Vesin et al., 2022; Zhang, 2024).

Recent advances in cognitive neuroscience and educational psychology have revealed significant relationships between programming education and cognitive development (Giannopoulou et al., 2020; Berssanette & De Francisco, 2022). Key findings indicate that programming activates specific brain regions, potentially enhancing the overall cognitive function (Giannopoulou et al., 2020; Silva et al., 2021). Diverse programming paradigms improve specific cognitive abilities, particularly abstract and spatial reasoning (Mansoor, 2021; Bubnic et al., 2024). Furthermore, programming education influences metacognitive skills that are crucial for problem-solving and SDL (Silva et al., 2021; Chen et al., 2024).

These cognitive developments bolster competencies delineated in the CBE framework. Enhanced abstract reasoning skills support problem-solving competencies, whereas improved metacognitive abilities facilitate SDL (Cunha et al., 2024; Etemi et al., 2024). However, several challenges remain. Individual differences substantially impact the effectiveness of programming learning (Hanafi et al., 2024; Navarro-Cota et al., 2024). Optimizing the cognitive load in programming education poses challenges, especially for novice learners (Morrison, 2013; Berssanette & De Francisco, 2022). Additionally, the degree to which programming-related cognitive skills are transferred to other domains remains debatable (Czerkowski & Lyman, 2015; Ibrahim Nnass et al., 2023).

Our pedagogical model systematically addresses these multifaceted challenges through a comprehensive integration of evidence-based approaches: technical proficiency development via carefully sequenced programming exercises (Assyne et al., 2022; Cunha et al., 2024); cognitive enhancement through strategically designed algorithmic thinking tasks (Saad & Zainudin, 2022; LEE & LEE, 2024); metacognitive advancement through structured reflective practice protocols (Silva et al., 2021; Mohammadi, 2024); and optimized cognitive load management through adaptive learning strategies tailored to individual student capabilities (Gligorea et al., 2023; Arigye, 2024).

Although CBE demonstrates theoretical potential, its long-term efficacy in programming education remains largely unexplored. While previous studies have attempted to improve programming education through various approaches, most have been limited to validating the short-term effects of a single teaching method (Luxton-Reilly et al., 2018; Medeiros et al., 2019). There is an urgent need to establish a systematic teaching model and verify its effectiveness in enhancing students' cognitive abilities and professional competencies through longitudinal research (Malhotra et al., 2023; Clear et al., 2020).

2.2 Computational thinking: synthesizing core competencies

CT continues to develop as a critical 21st-century skill, with recent research broadening its conceptual framework and assessment methodologies (Poulakis & Politis, 2021; Zhang & Specht, 2022). Our model incorporates CT as a bridge between cognitive development and practical programming competencies.

Key aspects of CT research include expanded CT frameworks that integrate ethical and social impact considerations (Smith et al., 2023), assessment tools that develop metrics to quantify CT skills and evaluate the effectiveness of educational interventions (Perez et al., 2018; Lu et al., 2022), and STEM integration, which fosters problem-solving and innovation capacities across disciplines (Zakaria & Iksan, 2020; Burbaite et al., 2018). However, significant challenges persist in the field of CT education. Establishing valid and reliable assessment tools for CT skills remains a significant hurdle (Lu et al., 2022). The extent of CT skill transferability across domains remains controversial (Czerkawski & Lyman, 2015; Ibrahim Nnass et al., 2023; Saba et al., 2023). Additionally, seamless incorporation of CT into diverse educational contexts presents ongoing challenges (Muñoz et al., 2023).

Our model addresses these challenges by offering a comprehensive framework that synthesizes CT with other key competencies and cognitive skills.

2.3 Self-directed learning: cultivating lifelong competency development

SDL has emerged as a crucial component of programming education, underscoring the need for lifelong learning in this rapidly evolving field (Boyer et al., 2008). Our model integrates SDL strategies to optimize the effectiveness of CBE and to facilitate ongoing cognitive and CT skill development.

Current research has focused on blended learning models that integrate online self-learning modules with personalized tutoring to enhance learning outcomes

(Faessler & Dahinden, 2023; Tirronen & Isomöttönen, 2011). Online resources leverage MOOCs and other platforms to foster SDL and develop self-regulatory skills (Onah et al., 2021). Additionally, adaptive technologies harness AI and big data to design personalized learning pathways (Zhang, 2024; Arigye, 2024). Empirical studies have yielded encouraging results in SDL for programming education. Mozeleus and Humble (2020) demonstrated that SDL and communities of practice can significantly alleviate the challenges in programming education. Polat et al. (2022) emphasized the positive impact of flipped learning models on students' autonomous learning capabilities in programming courses.

However, the implementation of SDL in programming education presents several challenges. These include varying levels of student preparedness and motivation for effective SDL (Onah et al., 2021), maintaining the quality and relevance of online learning materials (Etemi et al., 2024), and achieving an optimal equilibrium between providing guidance and fostering autonomy (Chen et al., 2024).

Our model addresses these challenges by integrating SDL strategies with other competency-based approaches, and provides a supportive framework for autonomous learning.

2.4 Synthesis and identification of research gap

Although extensive research has been conducted on CBE, cognitive development, CT, and SDL in programming education, there is a paucity of studies holistically integrating these elements into a comprehensive model. This gap is particularly pronounced in longitudinal studies that have investigated the combined impact of these approaches on students' programming competencies and cognitive development.

Our proposed integrated model aimed to address this research gap by amalgamating CBE, cognitive development strategies, CT frameworks, and the SDL approach with a coherent educational model. We established a framework for the longitudinal assessment of the model's impact on students' programming competencies and cognitive abilities. Furthermore, we address the challenges identified in each component area using a synergistic approach.

This literature review elucidates the potential for integrating CBE, cognitive development, CT, and SDL into programming education. Although each approach has its merits, its synergistic potential remains largely unexplored. Our proposed model endeavors to bridge this gap by offering a comprehensive framework for advancing programming education and cultivating crucial 21st-century skills.

3 Methodology

This section describes our research approach for investigating the effectiveness of a competency-based teaching model in enhancing students' cognitive abilities and programming competencies. We present a comprehensive overview of the research design, participant characteristics, implementation of the teaching model, data collection methods, and the analytical approaches employed in this 12-year longitudinal study.

3.1 Research design

We implemented a rigorous multiphase research design to develop, implement, and evaluate a competency-based programming education model throughout a 12-year longitudinal investigation (2011–2023). The methodological framework was deliberately designed to be iterative and adaptive, enabling continuous evidence-based refinement in response to technological advancements, changing student demographics, and unforeseen external challenges, including the significant educational disruptions caused by the COVID-19 pandemic.

This investigation proceeded through five sequential, interconnected phases that systematically built upon preceding outcomes. In the Initial Model Development Phase (2011–2013), we constructed a preliminary teaching framework through comprehensive systematic literature review (analyzing 127 relevant studies) combined with structured analysis of empirical teaching experiences, enabling the identification and validation of core competency indicators and corresponding assessment methodologies with high construct validity.

In the Early Implementation and Adjustment Phase (2014–2016), we conducted pilot implementations in selected cohorts and gathered feedback from the students and instructors for model refinement. The Model Optimization and Expansion Phase (2017–2019) involved systematic enhancements based on empirical evidence and expansion to additional cohorts and curricula, incorporating psychometric tools, such as (R.SPM) (Raven, 1938) to evaluate cognitive abilities. During the Comprehensive Implementation and In-depth Evaluation Phase (2020–2022), we scaled up the optimized model and conducted multidimensional data collection to track students' developmental trajectories longitudinally. Finally, the Analysis and Conclusion Phase (2023) involved a comprehensive meta-analysis of dual decennial data to evaluate the model's long-term effectiveness.

We adopted a mixed-methods approach incorporating a quasi-experimental design with longitudinal comparisons, survey instruments, semi-structured interviews, and standardized psychometric testing. This methodological triangulation allowed for a thorough evaluation of the model's effectiveness across different temporal periods, and the adaptation of pedagogical strategies in response to evolving educational environments.

Comprehensive methodological controls were implemented to address the potential limitations inherent in quasi-experimental designs. These included a cross-institutional validation framework established by Hunan Agricultural University and Jishou University in 2013 (with Xiangnan University joining in 2021), standardized assessment tools, uniform grading rubrics across institutions, and multiple data collection methods that combined quantitative assessments with qualitative data. Statistical controls included propensity score matching to balance group comparisons, sensitivity analyses to assess potential confounding variables, and effect-size calculations to determine practical significance.

3.2 Participants

The study included 4,051 undergraduate students enrolled in the C Programming course at Huaihua University, a regional tertiary institution in China, between the period 2011–2023. Annual cohorts ranged from 150 to 600 students, predominantly first-year undergraduates in computer-related fields of study. Participants entered the program with heterogeneous levels of programming proficiency, from complete novices to those with rudimentary coding experience.

The C Programming course encompassed 80 contact hours (32 h of theoretical instruction and 48 h of practical application), equivalent to four academic credits. All participants successfully completed the National College Entrance Examination (Gaokao) and fulfilled the university's matriculation criteria for computer-related programmes.

For a comparative analysis, we established three distinct groups to evaluate teaching efficacy during pandemic-induced educational disruptions: a pre-intervention control group (2013 and earlier) representing outcomes under traditional instruction, a post-intervention experimental group (2014–2020, 2023) representing outcomes under the competency-based teaching model, and a COVID-19 control group (2021–2022) to evaluate teaching efficacy during pandemic-induced educational disruptions.

From 2013 onwards, we established a joint examination mechanism with two tier-one universities (Hunan Agricultural University and Jishou University) to ensure the consistency and comparability of course delivery and assessment outcomes. From 2021, Xiangnan University has participated in this collaborative assessment framework.

All participants received comprehensive information regarding the study objectives and methodologies and informed consent was obtained from all participants. Participant confidentiality was safeguarded through data anonymization and secure data-management protocols.

3.3 Competency-based teaching model

Our competency-based teaching model was specifically designed to address the three core research questions identified in the introduction: enhancing learning outcomes and knowledge acquisition, developing cognitive abilities and computational thinking, and fostering self-directed learning abilities and professional competencies. By integrating multiple educational theories into a coherent framework, this model directly responds to persistent challenges in programming education: bridging the theory-practice gap, updating curriculum content, enhancing practical skills, cultivating innovative thinking, and increasing student engagement. Unlike traditional programming education approaches, which often focus on the isolated aspects of learning, our model offers a comprehensive solution that simultaneously addresses multiple dimensions of student development.

The theoretical foundation of our model integrates six educational theories, each addressing distinct, but interconnected aspects of programming education. This integration creates a comprehensive framework that enhances the learning outcomes through synergistic effects.

3.3.1 Theoretical framework

Our model is underpinned by constructivist learning theory (Vijayakumar & Pande, 2024), situated teaching theory (Tehrani et al., 2021), cooperative learning theory (Baligar et al., 2022), self-directed learning theory (Mohammadi, 2024), educational technology theory (Lei, 2017), and humanistic theory (Du, 2012). Constructivist learning theory emphasizes the active construction of knowledge by learners, promoting the development of higher-order thinking skills through practical and innovative projects. Situated teaching theory accentuates learning in authentic contexts, facilitating connections between abstract concepts and concrete applications through real-world programming tasks and case analyses. Cooperative learning theory focuses on peer interaction and collaboration, fostering social cognitive abilities through group projects and discussion. The self-directed learning theory emphasizes learners' autonomy in managing their learning processes and cultivating metacognitive abilities through autonomous learning tasks and reflective activities. The educational technology theory optimizes teaching processes through modern technological integration, thereby expanding students' cognitive resources through online learning platforms and programming tools. Humanistic theory emphasizes respect for individual differences and holistic development, fostering emotional and moral cognitive abilities through personalized guidance and education.

The integration of these theories provides a robust theoretical foundation for a model that collectively supports comprehensive enhancement of students' cognitive abilities. This integration manifests as a model structure with three dimensions, four integrations and five combinations.

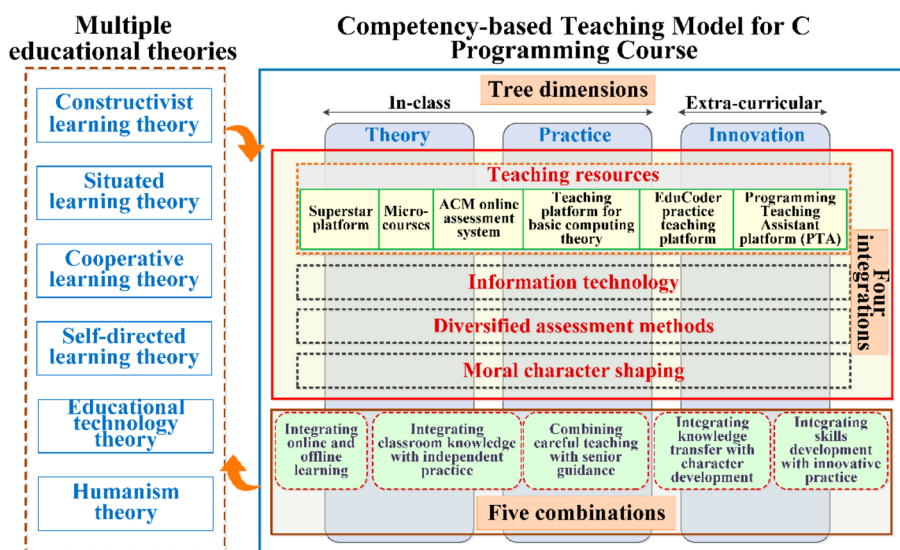
The innovative aspect of our theoretical framework lies in the synergistic integration of these six complementary theories, each of which addresses different, but interconnected, aspects of programming education. While previous approaches have typically employed one or two of these theories in isolation, our model leverages their combined strengths to create a more comprehensive learning environment. This integration is particularly effective in addressing contemporary programming education challenges because it simultaneously targets multiple learning dimensions: constructivist and situated learning theories address the theory-practice gap; educational technology theory responds to outdated curriculum content; cooperative learning and humanistic theories enhance student engagement; and self-directed learning theory promotes lifelong learning capabilities that are essential in the rapidly evolving field of programming. Table 1 illustrates how these six educational theories are integrated into our teaching model, highlighting their distinct focus areas, implementation strategies, and expected outcomes.

3.3.2 Model structure and components

As illustrated in Fig. 1, our competency-based teaching model integrates multiple theoretical foundations into a coherent framework designed to enhance students' cognitive and programming abilities. The figure depicts the hierarchical and interconnected nature of the model, and the six educational theories (left side) form the foundational layer that informs the three core dimensions (top), which are opera-

Table 1 Theoretical integration framework of the competency-based teaching model

Theory	Key focus	Implementation strategy	Expected outcome
Constructivist Learning	Active knowledge construction	Project-based learning activities	Enhanced problem-solving skills
Situated Learning	Contextual application	Industry-based projects	Improved practical competency
Cooperative Learning	Peer interaction	Team programming tasks	Developed team-work capabilities
Self-directed Learning	Learning autonomy	Personalized learning paths	Advanced SDL readiness
Educational Technology	Modern tool integration	Online platforms & tools	Enhanced digital literacy
Humanistic Theory	Holistic development	Ethics-integrated projects	Formed professional values

**Fig. 1** Competency-based teaching model for C programming course

tionalized through four integrations (middle) and implemented via five combinations (bottom). This layered structure creates a comprehensive system in which theoretical principles cascade into practical educational strategies, ultimately converging with the goal of cognitive enhancement through programming education.

The model is comprised of three core dimensions: theoretical, practical, and innovative. The theoretical dimension encompasses C-language fundamentals, algorithm-design principles, and data structures, thereby providing a cognitive foundation for programming expertise. The practical dimension involves hands-on programming exercises, project development, and problem-solving activities, which allow students to apply and reinforce their theoretical knowledge. The innovation dimension focuses on cultivating innovative thinking, encouraging novel algorithm designs, and exploring interdisciplinary applications to foster creative problem solving skills.

These three dimensions support each other and form a spiraling process during cognitive development. For instance, students learn about sorting algorithms theoretically, implement them in practice, and innovate by optimizing algorithms for specific use cases, thereby iteratively deepening their understanding and skills.

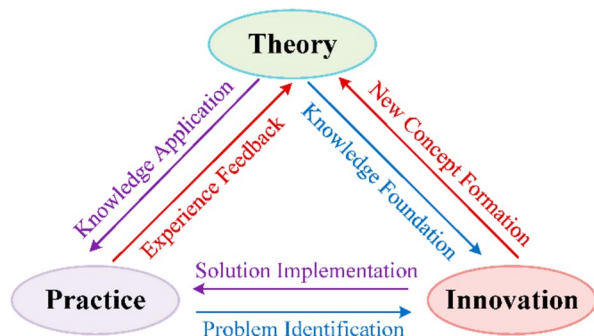
Figure 2 depicts the dynamic interrelationships among these three dimensions, showing how they interact to create a synergistic learning environment that promotes comprehensive skill development.

This model incorporates four key integrations that synergistically enhance cognitive development. These integrations function as interconnected systems rather than isolated components. Resource integration provides a comprehensive knowledge base through multiple platforms (superstar platforms, micro-courses, ACM online assessment systems, etc.), supporting the development of declarative and procedural knowledge structures. Information technology integration complements this by enhancing learning experiences through digital tools that expand information-processing capabilities. Diversified assessment methods work in concert with previous integration methods by providing feedback mechanisms that promote metacognitive awareness and self-regulation. Moral character-shaping integration weaves ethical considerations throughout other components, ensuring that cognitive development encompasses professional responsibility and ethical decision making. This coordinated integration creates a learning ecosystem, in which each component reinforces and amplifies the effects of others.

Additionally, the model features five combinations: integrating online and offline learning provides flexible learning opportunities, integrating classroom knowledge with independent practice reinforces learning, combining careful teaching with senior guidance provides multilevel support, integrating knowledge transfer with character development ensures balanced growth, and integrating skill development with innovative practices to cultivate practical abilities and creative thinking.

Each dimension, integration, and combination in our model targets specific aspects of cognitive development. The theoretical dimension primarily enhances abstract reasoning and conceptual understanding, the practical dimension strengthens procedural knowledge and problem-solving abilities, and the innovation dimension fosters creative thinking and cognitive flexibility. When these elements interact in the educational process, they create multiple pathways for cognitive enhancement, addressing

Fig. 2 Integration of three dimensions in the competency-based teaching model



different learning styles and cognitive preferences, while building a comprehensive set of programming competencies.

These elements collectively enhance students' cognitive abilities through multiple stimuli, deep processing, situated learning, social interaction, reflection, metacognition, and technological enhancement.

3.3.3 Evolution and refinement of the model

Since its inception in 2011, this model has undergone iterative development and refinement in five distinct phases. The Initial Development Phase (2011–2013) focused on integrating theoretical and practical dimensions and introducing rudimentary online learning components. The Early Implementation and Adjustment Phase (2014–2016) incorporated the innovation dimension and expanded the utilization of educational technology. The Refinement Phase (2017–2019) introduced moral character-shaping integration and enhanced the senior guidance components. The Comprehensive Implementation Phase (2020–2022) augmented online learning components in response to the COVID-19 pandemic and intensified industry-relevant project integration. The Continuous Improvement Phase (from 2023 onwards) features data-driven adjustments based on rigorous analysis of student performance metrics and industry feedback.

Throughout its evolution, the model has consistently aimed to address the pervasive challenges in programming education, including knowledge fragmentation, superficial learning, and the diverse cognitive needs of heterogeneous student populations. Key improvements include the enhanced integration of theoretical concepts with practical applications, increased emphasis on innovative problem solving, progressive incorporation of technology-enhanced learning, greater focus on ethical considerations, and adaptation to diverse learning needs through personalized learning paths.

The evolutionary development of this model demonstrates its adaptability to changing educational needs and technological advancements. Through continuous refinement based on empirical evidence and stakeholder feedback, we created a teaching framework that not only addresses the immediate challenges in programming education, but also provides a sustainable approach to cognitive enhancement. By systematically developing students' theoretical understanding, practical skills, and innovative capabilities, the model creates an educational environment in which cognitive development occurs naturally through meaningful engagement with programming concepts and practices.

This approach aligns with our research objectives of enhancing students' learning outcomes, developing their cognitive abilities, and fostering self-directed learning competencies, thus establishing the foundation for the empirical results presented in subsequent sections.

3.4 Implementation strategies

We developed and implemented several pedagogical strategies based on a competency-based teaching model, with the primary objective of effectively integrating the dimensions of theory, practice, and innovation.

3.4.1 Integrated learning approach

We engineered a multidimensional learning paradigm that deliberately integrates theoretical knowledge acquisition, practical skill application, and innovative exploration through carefully orchestrated instructional sequences throughout the curriculum. This pedagogical innovation is operationalized through two complementary approaches: our evidence-based spiral learning model and our project-driven learning strategy.

The spiral learning model implements a tripartite cyclical structure where concepts are revisited with increasing complexity. For each instructional unit, this cycle includes: theoretical acquisition of new concepts and principles; practical application through structured programming exercises that consolidate knowledge; and innovative exploration that challenges students to develop novel solutions based on existing knowledge. For example, in teaching pointers, students engage with pre-class videos explaining pointer concepts, complete pointer-related exercises on an Online Judge (OJ) platform, and design small-scale projects that utilize pointers to optimize program performance.

Complementing this approach, our project-driven learning strategy contextualizes abstract principles within authentic programming challenges. Weekly theoretical courses are directly correlated with project requirements, thereby ensuring their relevance and applicability. Students apply learned theories to solve practical problems within the project framework and incorporate innovative elements such as algorithm optimization or feature augmentation.

3.4.2 Diverse pedagogical methodologies

Diverse pedagogical methodologies have been employed to support our integrated approach. These included flipped classroom techniques in which students engaged with pre-recorded lectures before class discussions, interactive classroom discussions in small groups to deepen the understanding of theoretical concepts, concept mapping to visualize relationships between interconnected concepts, progressive programming tasks of incrementally increasing complexity, real environment simulation projects designed to mimic authentic work environments, an OJ system providing immediate feedback, open-ended projects encouraging innovative applications, on-campus programming competitions, and research training opportunities for high-performing students.

We implemented structured cooperative learning mechanisms through 4–5 member teams with complementary skills, systematic role rotation, and progressive tasks. Multilevel interaction patterns were fostered through in-class discussions, out-of-

class project collaboration, and code review practices, supported by online collaborative programming environments.

3.4.3 Assessment framework

Our assessment system was meticulously designed to evaluate student performance across theoretical understanding, practical implementation capability, and innovative problem-solving capacity. We implemented a validated multidimensional assessment protocol comprising weighted components: criterion-referenced theoretical examinations (40%), competency-based programming assignments and project implementation (30%), rubric-guided open-ended problem solving and innovative project assessment (20%), and structured peer evaluation protocols (10%).

This balanced approach ensures comprehensive measurement of both technical proficiency and higher-order cognitive skills. Regular reflection sessions facilitated students' integration of theoretical, practical, and innovative experiences, with instructors using these reflections to inform pedagogical adjustments.

3.4.4 Implementation challenges and solutions

In implementing this integrated approach, we encountered several challenges and developed the corresponding mitigation strategies. Student adaptability issues were addressed through a peer mentorship system in which senior students provided guidance to junior cohorts. Teaching resource limitations was overcome through a tripartite approach involving self-development, external importation, and collaborative creation of resources. Assessment complexities were addressed by using a diversified assessment system to evaluate outcomes, problem-solving processes, and collaborative abilities.

We maintained continuous monitoring and evaluation protocols to assess the efficacy of these strategies, and made data-driven adjustments based on feedback from both students and instructors, ensuring continuous improvement and alignment with evolving educational and industry standards.

3.5 Data collection and analysis

To evaluate the efficacy of the competency-based teaching model comprehensively, we implemented a multidimensional approach to data collection and analysis, covering various aspects of student performance and development.

3.5.1 Academic performance assessment

We collected data on theoretical knowledge through examinations, practical project evaluations, OJ system performance metrics, and classroom engagement indices. These components collectively contribute to the evaluation of the course objectives achievements (Yang, 2024). Additionally, we administered concept tests and student self-assessment questionnaires, and collected data through collaborative examina-

tion mechanisms with partner institutions (Authement et al., 2023; Courtis & Packer, 1920).

3.5.2 Cognitive ability assessment

We employed a comprehensive approach that integrates qualitative and quantitative assessment methods (Latha & Mohanamani, 2022). Our assessment battery included the R.SPM to evaluate general cognitive ability, the CT Test (Zhang et al., 2024) to assess CT skills, complex programming tasks to measure problem-solving abilities (Bubnic et al., 2024), open-ended projects to evaluate creative thinking (LEE & LEE, 2024), and metacognitive ability questionnaires to assess self-reflective learning processes.

3.5.3 Skills and competency evaluation

We documented students' performance in programming competitions (Zhang et al., 2021), tracked participation in innovative activities (Chen et al., 2013), projects, publications, and patent applications; and conducted code reviews and technical report assessments. For teamwork and communication skills, we employed the Team Project Evaluation Scale (Chyung et al., 2017), peer evaluation systems (Chen, 2010), Communication Competence Self-Report Scale (CCSS) (Croucher et al., 2019), and Team Role Diversity Index (Altiner & Ayhan, 2018). We evaluated SDL abilities using the SDL Readiness Scale (SDLRS) (Long & Agyekum, 1983), Time Management Questionnaire (TMQ) (Britton & Tesser, 1991), Self-Regulated Learning Questionnaire (SRLQ) (Öz & Şen, 2018), and Metacognitive Awareness Inventory (MAI) (Harrison & Vallin, 2018).

For ethical character assessment, we used the Defining Issues Test-2 (DIT-2) (Gungordu et al., 2023), custom Ethical Decision-Making Scale (EDM) (Mumford et al., 2008), Professional Ethics Knowledge Test (Miñano et al., 2017), Ethical Behavior Observation Scale (Cardy & Selvarajan, 2006), and an automated plagiarism detection system (Tahaei & Noelle, 2018). All the instruments were validated for use in the Chinese context.

3.5.4 Data analysis methods

We employed diverse statistical techniques, including descriptive statistics for data summarization (Pyzdek, 2021), inferential statistics (paired t-tests and ANOVA) for hypothesis testing (Sanders et al., 2019; Lakens, 2013), correlation analyses to examine relationships between variables (Cohen et al., 2002), multiple regression analysis for predictive modeling (Cohen, 1968), mediation analyses to identify intervening variables (Kong & Wang, 2020), trend and comparative analyses, longitudinal analysis to track changes over time, thematic analysis for qualitative data interpretation (Braun & Clarke, 2006), and effect size calculations to quantify observed effects (Fritz et al., 2012).

All statistical analyses were performed using IBM SPSS Statistics (version 27.0; IBM Corp., Armonk, NY, USA), strictly adhering to APA reporting guidelines with a significance level of $\alpha=0.05$.

3.6 Ethical considerations

We implemented a comprehensive framework of ethical safeguards, emphasizing compliance with Chinese regulations while adhering to international best practices. Written informed consent was obtained from all participants, emphasizing the voluntary nature of their participation in accordance with China's Personal Information Protection Law. We implemented robust data encryption and secure storage methods that adhered to China's cybersecurity and data security laws.

The study was approved by the university's Institutional Review Board, ensuring compliance with the established ethical guidelines and relevant Chinese regulations. We appointed a dedicated data protection officer, conducted regular risk assessments, and established data classification and protection measures as required by Chinese laws.

This study was designed to benefit all student cohorts equally with no identified foreseeable risks. A comprehensive debriefing process was conducted upon completion of the study. All the research materials were culturally appropriate and linguistically accurate in the Chinese context. We have committed to the transparent reporting of research methods and findings, including limitations and potential biases, in accordance with the principles of scientific integrity and open science.

These measures collectively ensured a robust ethical framework for our research by balancing the local regulatory requirements with global best practices in educational research ethics.

4 Results

This section presents the findings of our 12-year longitudinal study, which examined the effectiveness of a competency-based teaching model in enhancing students' cognitive abilities and programming competencies. The results were organized to address three research questions concerning learning outcomes, cognitive development, and SDL abilities.

4.1 Impact on academic performance and knowledge acquisition

This longitudinal study conducted a comparative analysis of student learning outcomes before and after implementation of a competency-based teaching model in a C programming course at Huaihua University. Data collected until 2013 represented outcomes under the traditional instructional paradigm (pre-intervention control group), whereas data from 2014–2020 and 2023 constituted the post-intervention experimental group. Data from 2021–2022 were analyzed separately to assess teaching efficacy during the COVID-19 pandemic.

Table 2 Joint examination paper analysis from 2013 to 2023

Year	Participants	Difficulty		Differentiation		Reliability	
		Theory	Practice	Theory	Practice	Theory	Practice
2013	816	0.612	0.491	0.532	0.501	0.721	0.782
2014	827	0.751	0.602	0.473	0.401	0.824	0.667
2015	862	0.666	0.639	0.425	0.615	0.685	0.701
2016	968	0.667	0.738	0.441	0.427	0.702	0.718
2017	963	0.608	0.660	0.491	0.417	0.715	0.705
2018	1075	0.706	0.617	0.441	0.614	0.792	0.727
2019	1158	0.729	0.559	0.468	0.526	0.856	0.801
2020	1217	0.702	0.678	0.526	0.623	0.849	0.823
2021	1452	0.472	0.537	0.643	0.702	0.784	0.814
2022	1579	0.437	0.511	0.631	0.694	0.798	0.823
2023	1762	0.525	0.508	0.649	0.682	0.801	0.809

Note: Difficulty, discrimination, and reliability coefficients were calculated using the standard methods for educational measurements

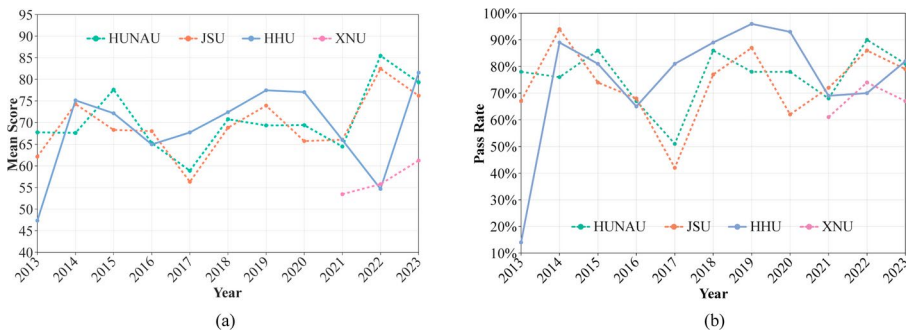


Fig. 3 Longitudinal analysis of theoretical examination performance (2013–2023): (a) mean scores and (b) pass rates

4.1.1 Enhancement in academic performance

Building on the collaborative examination mechanism established in 2013, we refined this assessment framework to ensure the validity and comparability of course instruction across participating institutions. This joint assessment protocol has involved Huaihua University (HHU), Hunan Agricultural University (HUNAU), Jishou University (JSU), and Xiangnan University (XNU) since 2021.

The consortium administered unified examinations using a common assessment platform (Programming Teaching Assistant, PTA) with examination content aligned with professional accreditation standards. Table 2 presents the analysis of examination papers from 2013 to 2023, showing the carefully controlled difficulty, discrimination, and reliability indices.

Figures 3 and 4 present a comparative analysis of the examination results, demonstrating a significant improvement in students' knowledge mastery and academic performance after the introduction of the competency-based teaching model.

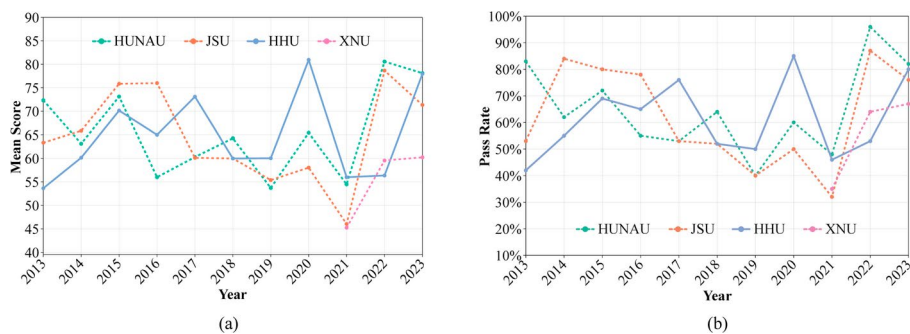


Fig. 4 Longitudinal analysis of practical examination performance (2013–2023): (a) mean scores and (b) pass rates

Table 3 Longitudinal analysis of performance metrics for assignments, practical projects, and extracurricular practical projects (2011–2023)

Year	Assignments			Practical projects			Extracurricular practical projects		
	Average marks	Completion rates	Excellence rates	Average marks	Completion rates	Excellence rates	Average marks	Completion rates	Excellence rates
2011	72.35	85.12%	28.52%	84.12	87.13%	51.32%	76.45	80.45%	28.47%
2012	71.68	86.03%	28.73%	83.45	87.22%	51.89%	78.65	81.68%	29.52%
2013	73.19	87.19%	29.23%	82.69	88.37%	55.72%	79.96	82.59%	36.25%
2014	81.52	91.41%	36.23%	89.06	92.34%	61.12%	82.56	87.45%	38.26%
2015	84.19	95.42%	38.45%	96.21	94.67%	74.04%	89.37	89.21%	45.24%
2016	85.31	96.67%	39.18%	96.48	95.03%	77.25%	91.21	89.89%	45.53%
2017	87.35	97.48%	40.05%	95.98	98.02%	79.57%	90.38	89.07%	44.76%
2018	88.27	98.12%	40.19%	96.23	98.34%	84.15%	92.34	90.23%	58.58%
2019	90.21	99.52%	41.27%	93.54	98.54%	80.01%	91.42	90.09%	60.15%
2020	89.48	98.56%	40.63%	93.77	96.34%	81.05%	90.75	90.27%	61.14%
2021	89.73	98.97%	40.42%	94.45	97.25%	82.25%	90.35	89.25%	60.23%
2022	88.69	96.41%	39.69%	90.35	94.89%	79.76%	87.52	88.05%	56.87%
2023	89.23	99.78%	41.56%	96.27	97.41%	83.22%	91.56	91.14%	62.04%

Note: Assignment data were extracted from the Superstar and Computer Basic Theory Teaching Platform; practical training project data were sourced from the PTA and Computer Basic Theory Teaching Platform; extracurricular practical project data were obtained from the EduCoder Platform

As evidenced in Figs. 3 and 4, the implementation of the competency-based teaching model corresponded with statistically significant improvements in both theoretical and practical examination performance metrics. During the baseline assessment in 2013, when our institution employed conventional pedagogical methodologies, students demonstrated substantially lower mean scores and pass rates in both theoretical and practical examinations compared to tier-one universities ($p < 0.01$, Cohen's $d > 0.8$, indicating a large effect size).

From 2014 onwards, following the implementation of the competency-based teaching model, student performance significantly improved. Mean scores and pass rates demonstrated a consistent upward trajectory in some intervals, surpassing those

of tier-one institutions ($p < 0.05$, Cohen's $d > 0.6$). However, the COVID-19 pandemic in 2021–2022 caused disruptions in teaching and learning modalities, resulting in a decline in outcomes during this period, although they remained significantly elevated relative to pre-reform levels ($p < 0.05$, Cohen's $d > 0.4$).

4.1.2 Enhancement in knowledge mastery

To evaluate students' knowledge acquisition and retention comprehensively, we analyzed their completion rates, mean scores, and excellence rates across assignments, practical projects, and extracurricular practical projects. Table 3 presents the results for 2011–2023.

A repeated-measures Analysis of Variance (ANOVA) revealed statistically significant improvements in completion rates, mean scores, and excellence rates across all three assessment modalities following the implementation of the competency-based teaching model ($p < 0.01$, partial $\eta^2 > 0.25$). Post-hoc pairwise comparisons using Bonferroni-adjusted t-tests demonstrated that the post-reform period (2014–2020, 2023) exhibited significantly higher values than the pre-reform period (2011–2013) across all assessment types ($p < 0.01$, Cohen's $d > 0.8$).

Statistical analysis of longitudinal data revealed significant correlations between specific curriculum adjustments and student performance metrics. The integration of theoretical and practical components showed the strongest correlation with improved learning outcomes ($\beta = 0.42$, $p < 0.001$), followed by the addition of innovation-focused components ($\beta = 0.38$, $p < 0.001$). Multiple regression analysis indicated that these curriculum refinements collectively explained 43% of the variance in student achievement ($R^2 = 0.43$, $F(3, 347) = 87.23$, $p < 0.001$).

4.1.3 Effectiveness of theoretical framework integration

Analysis of our theoretical framework revealed significant contributions from each theoretical component, as summarized in Table 4.

Multiple regression analysis demonstrated significant synergistic effects, with the combined framework explaining 43% of the variance in learning outcomes ($R^2 = 0.43$, $F(3, 347) = 87.23$, $p < 0.001$), while individual theories contributed between 15 and 25% each. This indicates a synergistic gain ($\Delta R^2 = 0.18$, $p < 0.001$) from the integrated approach.

Path analysis revealed specific theoretical contributions through knowledge construction pathways (academic performance: $\beta = 0.42$, $p < 0.001$; problem-solv-

Table 4 Quantitative analysis of theoretical framework components

Theory	Measured impact	Statistical significance	Effect size
Constructivist Learning	$\beta = 0.35$	$p < 0.001$	$d = 0.84$
Situated Learning	$\beta = 0.28$	$p < 0.010$	$d = 0.76$
Cooperative Learning	$\beta = 0.31$	$p < 0.001$	$d = 0.79$
Self-directed Learning	$\beta = 0.29$	$p < 0.010$	$d = 0.72$
Educational Technology	$\beta = 0.26$	$p < 0.010$	$d = 0.68$
Humanistic Theory	$\beta = 0.22$	$p < 0.050$	$d = 0.65$

ing skills: $\beta=0.38$, $p<0.001$), practical application pathways (skill development: $\beta=0.38$, $p<0.001$; project completion: $\beta=0.35$, $p<0.001$), and innovation development pathways (creative thinking: $\beta=0.35$, $p<0.001$; novel solution generation: $\beta=0.32$, $p<0.001$).

4.2 Enhancement in curriculum attainment

In response to China's accession to the Washington Accord in 2016, Huaihua University revised its educational programs to align with the international engineering education standards. We conducted an in-depth evaluation to assess the impact of implementing a competency-based teaching model on the attainment of C programming course objectives.

The revised curriculum incorporated a comprehensive set of assessment protocols designed to measure students' mastery of the theoretical constructs and practical applications. Table 5 presents the longitudinal analysis of course objective attainment from 2018 to 2023.

Table 5 shows consistent improvements across both theoretical and practical domains. The weighted average of theoretical course attainment (coefficient 0.5) and practical course attainment (coefficient 0.5) shows an overall positive trend, with composite course objective attainment increasing from 0.809 in 2018 to 0.838 in 2023, despite minor fluctuations potentially attributable to the COVID-19 pandemic.

Through rigorous data analysis, it is evident that this innovative pedagogical approach has been instrumental in promoting the achievement of course objectives, and has exerted a profound impact on the development of students' cognitive faculties. The model emphasizes the centrality of cognitive skill development in classroom instruction, effectively enhancing students' analytical reasoning, problem-solving aptitude, critical thinking capabilities, creative cognition, and overall cognitive competencies.

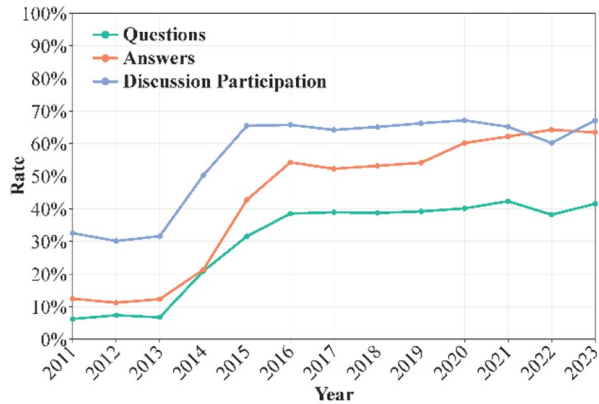
4.3 Enhancement in student interest and cognitive engagement

We comprehensively investigated the efficacy of a competency-based teaching model for stimulating student interest and enhancing cognitive engagement. The analysis focused on three key engagement indicators: student-initiated inquiries, student-pro-

Table 5 Statistical data on the attainment of course objectives, 2018–2023

Year	Theoretical course attainment (0.5)				Practical course attainment (0.5)			Course objective attainment
	Objective 1 (0.4)	Objective 2 (0.4)	Objective 3 (0.2)	Attainment	Objective 1 (0.7)	Objective 2 (0.3)	Attainment	
2018	0.841	0.675	0.859	0.778	0.843	0.831	0.839	0.809
2019	0.834	0.670	0.764	0.754	0.826	0.803	0.819	0.787
2020	0.806	0.853	0.818	0.827	0.906	0.894	0.902	0.865
2021	0.807	0.695	0.831	0.767	0.821	0.798	0.814	0.791
2022	0.772	0.735	0.752	0.753	0.807	0.788	0.801	0.777
2023	0.812	0.727	0.847	0.785	0.899	0.871	0.891	0.838

Fig. 5 Longitudinal analysis of student engagement metrics pre- and post-implementation of the competency-based teaching model



vided responses, and participation in academic discourse. The results are shown in Fig. 5.

Figure 5 illustrates that the implementation of the competency-based teaching model resulted in statistically significant enhancements in student-instructor and peer-to-peer interactions within the classroom setting. A marked increase in the proportion of student-generated questions was observed, indicating heightened intellectual curiosity and a willingness to engage with complex subject matter. The proportion of student-generated answers exhibited notable increases, suggesting an augmentation of both content mastery and self-efficacy. Additionally, substantial increases in participation during academic discussions reflect an enhanced active involvement in collaborative knowledge construction.

The data reveal a clear upward trajectory in engagement metrics from the pre-implementation period (2011–2013) to post-implementation years (2014–2020). Notably, even during the challenges posed by the COVID-19 pandemic (2021–2022), the model demonstrated robust resilience in maintaining student interest and engagement, albeit with expected fluctuations.

Further statistical analysis revealed that concomitant with increased engagement, students exhibited significant improvements in multifaceted cognitive abilities. The observed increase in student-initiated inquiries and active participation in academic discourse suggests enhanced critical thinking, analytical reasoning, and metacognitive skills.

4.4 Enhancement of students' innovative capacity and practical proficiency

The competency-based teaching model had a significantly positive impact on students' cognitive faculties, practical skills, and innovation capabilities. This enhancement is evidenced by the students' increased engagement and success in various innovative practice projects that demand comprehensive knowledge applications and multifaceted abilities.

To comprehensively assess students' knowledge application, problem-solving acumen, and innovative design capabilities, we employed various evaluation criteria including programming competitions, innovation projects, publications, and intellec-

Fig. 6 Longitudinal analysis of programming competition awards

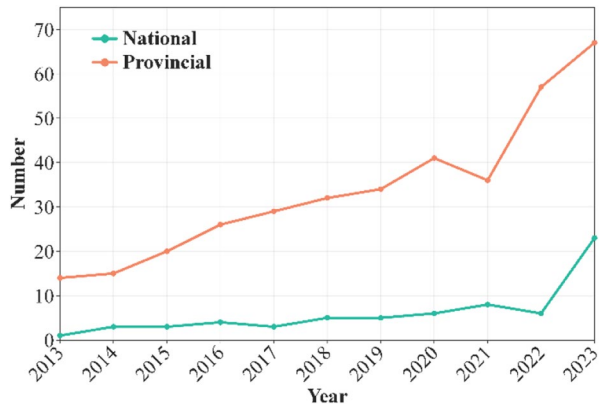
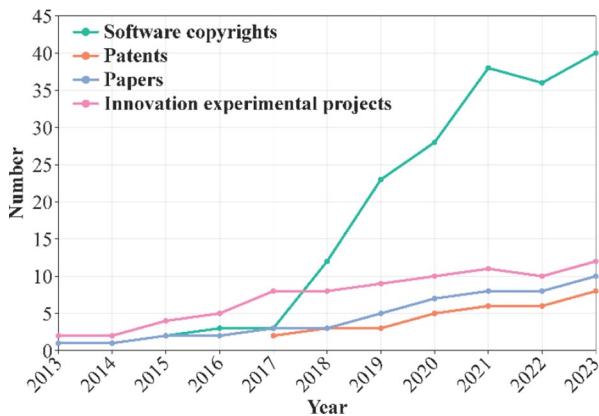


Fig. 7 Comparative analysis of students' innovation capabilities pre- and post-implementation of the competency-based teaching model



tual property applications. A longitudinal analysis of these metrics revealed significant improvements across all the dimensions.

Students mentored by the course team garnered 438 awards in various prestigious programming competitions, including the China Collegiate Computing Contest, China Collegiate Programming Contest, ACM-ICPC Asia Regional Contest, Hunan Province Collegiate Programming Contest, and the CCF Collegiate Computer Systems & Programming Contest. As illustrated in Fig. 6, the number of student awards demonstrated a consistent annual upward trajectory, even during the COVID-19 pandemic.

The course team successfully guided students in securing 26 national-and 55 provincial-level student innovation experimental projects. Cumulatively, the team facilitated the application of 185 software copyrights and 33 patents alongside the publication of 50 academic papers. Figure 7 illustrates the consistent upward trend in these achievements, with a slight decline during the COVID-19 pandemic attributable to disruptions in face-to-face mentorship.

These outstanding achievements provide empirical evidence of notable improvements in students' analytical thinking, problem-solving capabilities, creative cognition, and ability to synthesize and apply knowledge across diverse contexts.

Table 6 Comparative analysis of team project and communication metrics (2014 vs. 2023)

Measure	2014 score	2023 score	Statistical test result
Team Project Evaluation Scale	3.2 (SD=0.8)	4.1 (SD=0.6)	Paired sample <i>t</i> -test: <i>t</i> (305)=15.7, <i>p</i> <0.001, Cohen's <i>d</i> =0.84
Peer Evaluation	7.2/10 (SD=1.2)	8.6/10 (SD=0.9)	Independent sample <i>t</i> -test: <i>t</i> (305)=18.3, <i>p</i> <0.001, Cohen's <i>d</i> =0.95
CCSS	62/100 (SD=11)	78/100 (SD=9)	Paired sample <i>t</i> -test: <i>t</i> (305)=22.4, <i>p</i> <0.001, Cohen's <i>d</i> =1.19
Team Role Diversity Index	0.58 (SD=0.14)	0.73 (SD=0.11)	Independent sample <i>t</i> -test: <i>t</i> (305)=7.6, <i>p</i> <0.001, Cohen's <i>d</i> =0.79

Table 7 Longitudinal analysis of SDL and self-management skills (2014 vs. 2023)

Measure	2014 score	2023 score	Change	Statistical significance
SDLRS	214.3 (SD=22.7)	239.8 (SD=18.5)	+11.9%	<i>t</i> (305)=19.6, <i>p</i> <0.001, <i>d</i> =1.06
TMQ	3.2 (SD=0.6)	4.1 (SD=0.4)	+28.1%	<i>t</i> (305)=22.3, <i>p</i> <0.001, <i>d</i> =1.21
SRLQ	3.4 (SD=0.5)	4.3 (SD=0.3)	+26.5%	<i>t</i> (305)=28.7, <i>p</i> <0.001, <i>d</i> =1.55
MAI	185.3 (SD=24.6)	221.7 (SD=18.9)	+19.6%	<i>t</i> (305)=23.1, <i>p</i> <0.001, <i>d</i> =1.26

4.5 Enhancement of students' teamwork and communication skills

We employed multiple assessment methods to quantitatively evaluate improvements in students' teamwork and communication skills, including the Team Project Evaluation Scale, peer evaluation systems, CCSS, and Team Role Diversity Index. Table 6 presents the results of our analysis.

The analysis of cooperative learning outcomes revealed significant improvements in both team capabilities and learning effectiveness. Team performance showed a marked increase in communication competence (16-point increase, *p*<0.001), role adaptability, and collaborative problem solving. The learning outcomes demonstrated higher quality deliverables, improved knowledge sharing, and enhanced innovation through collaboration.

Despite the limitations of face-to-face interaction during the COVID-19 pandemic, the use of online collaboration platforms has enabled students to continue developing these crucial skills. These results demonstrate the effectiveness of the competency-based teaching model in fostering students' comprehensive development and preparing them for future careers.

4.6 Enhancement of SDL and self-regulatory competencies

Our longitudinal analysis, spanning from 2014 to 2023, revealed significant improvements in students' SDL and self-regulatory competencies. We employed multiple assessment methods, including the SDLRS, TMQ, SRLQ, and MAI. Table 7 presents the results of our analysis.

All standardized assessments showed statistically significant improvements, with the SRLQ demonstrating the largest effect size (*d*=1.55). Notably, the "metacognitive self-regulation" subscale of the SRLQ increased from 3.1 to 4.4 on a 5-point Likert scale.

Additional key findings included a substantial increase in the proportion of students exhibiting high SDL readiness (from 49.7 to 78.3%), a significant increase in the time allocated to self-directed study materials (68% increase), voluntary practice problems (112% increase), and goal achievement rates (from 61 to 87%). The most substantial gains in metacognitive awareness were observed in the “planning” and “monitoring” subscales of the MAI.

These findings provide robust empirical evidence for the efficacy of our competency-based teaching model in enhancing students’ SDL and self-regulatory competencies. The consistency of enhancement across diverse measures suggests that the model’s emphasis on autonomous learning and cognitive self-regulation was highly effective even during the COVID-19 pandemic.

4.7 Enhancement of ethical competence and axiological development

Our longitudinal analysis from 2014 to 2023 revealed significant improvements in the ethical competence and value systems of students. We employed a mixed-method approach that combined both quantitative and qualitative assessments. Table 8 presents the results of the study.

Key findings include a substantial increase in the proportion of students in the post-conventional moral reasoning stage (from 23 to 41%), significant improvements in EDM competencies (particularly in “recognition of ethical issues” and “EDM process” subscales), and a marked decrease in detected plagiarism rates (from 8.7 to 2.1% of submissions, $\chi^2(1, N=6824)=178.3, p<0.001$, Cramer’s $V=0.16$).

Qualitative data corroborated the quantitative results, revealing several key themes: increased awareness of the ethical implications of technological development, a heightened sense of social responsibility, and an enhanced capacity to navigate complex ethical dilemmas in professional contexts. Students frequently articulated a deeper understanding of the ethical dimensions of their work, demonstrating their ability to anticipate and address potential ethical challenges in terms of technological development and deployment.

Notably, even during the COVID-19 pandemic, the implementation of online case studies and moral scenario simulations successfully cultivated students’ cognitive empathy and moral reasoning abilities. These improvements across multiple measures suggest that integrating ethical considerations into our competency-based curriculum is highly effective for fostering ethical competence and technical skills.

Table 8 Quantitative measures of ethical competence and axiological development (2014 vs. 2023)

Measure	2014 score	2023 score	Change	Statistical significance
DIT-2 P-score	35.2 (SD=8.7)	44.6 (SD=7.2)	+26.7%	$t(305)=16.4, p<0.001, d=0.91$
EDM Competence	3.2 (SD=0.6)	4.3 (SD=0.4)	+34.4%	$t(305)=25.7, p<0.001, d=1.41$
Professional Ethics Knowledge	62% (SD=11%)	84% (SD=8%)	+35.5%	$t(305)=29.3, p<0.001, d=1.59$
Observable Ethical Behavior	3.4 (SD=0.7)	4.5 (SD=0.4)	+32.4%	$t(305)=8.6, p<0.001, d=1.33$

Table 9 Longitudinal comparison of students' mean R.SPM Raw scores (2011–2023)

Year	Number of test participants	Proportion (%)	Average raw score	t-values	p-values
2011	135	60.00%	50.12	1.012	0.001
2012	156	62.90%	49.43	1.205	0.001
2013	161	60.75%	50.18	1.309	0.002
2014	166	63.85%	54.23	1.410	0.000
2015	181	67.04%	55.62	1.034	0.001
2016	170	60.71%	55.43	1.508	0.000
2017	190	64.41%	56.28	1.421	0.002
2018	204	68.00%	55.39	1.392	0.000
2019	201	63.81%	54.76	1.207	0.001
2020	224	68.92%	56.12	1.169	0.000
2021	213	61.74%	55.71	1.385	0.000
2022	234	65.00%	56.09	1.437	0.001
2023	351	62.34%	56.52	1.326	0.001

4.8 Enhancing of students' cognitive abilities

This section presents a comprehensive analysis of the impact of our competency-based teaching model on students' cognitive abilities, focusing on the results of a longitudinal study conducted between 2011 and 2023.

Longitudinal analysis of R.SPM scores revealed a consistent upward trajectory in students' cognitive abilities following implementation of the competency-based model in 2014. Table 9 quantifies this enhancement, documenting an increase in mean R.SPM scores from 50.12 (SD=8.7) in 2011 to 56.52 (SD=7.2) in 2023, representing a substantial improvement of 6.40 points (12.8%) over the study period with consistently reduced variance. Paired sample *t*-tests confirmed this improvement was statistically significant ($t(350)=8.76, p<0.001$), with a medium to large effect size (Cohen's $d=0.68$).

Our analysis revealed significant positive correlations between R.SPM score improvements and various academic outcomes: theoretical exam performance ($r=0.62, p<0.001$), practical project execution ($r=0.58, p<0.001$), and engagement in innovative projects ($r=0.45, p<0.01$).

Multiple regression analysis identified the following key predictors of R.SPM score improvement: classroom engagement ($\beta=0.35, p<0.001$), practical project completion ($\beta=0.28, p<0.01$), and innovative activity participation ($\beta=0.22, p<0.05$). This model accounted for 43% of the variance in R.SPM scores ($R^2=0.43, F(3, 347)=87.23, p<0.001$).

Further analysis revealed strong positive correlations between improvements in R.SPM scores and domain-specific cognitive skills: CT proficiency ($r=0.71, p<0.001$) and problem-solving capability ($r=0.67, p<0.001$). Mediation analyses revealed that both CT skills (indirect effect=0.24, 95% CI [0.18, 0.31]) and problem-solving abilities (indirect effect=0.19, 95% CI [0.14, 0.25]) partially mediated the relationship between the R.SPM score improvement and overall academic performance.

Additional analyses revealed moderate positive correlations between R.SPM scores and innovation capability ($r=0.65$, $p<0.001$), SDL proficiency ($r=0.59$, $p<0.001$), and collaborative competence ($r=0.51$, $p<0.001$). These findings illuminate the multifaceted nature of cognitive enhancement within our pedagogical model, suggesting that improvements in general cognitive ability are associated with enhancements in various key competencies of programming education.

4.9 Longitudinal evaluation: preliminary findings

Extending beyond immediate educational outcomes, this section presents preliminary findings from our systematic longitudinal evaluation examining the sustained post-graduation impact of the competency-based teaching model. This critical evaluation component employs multiple data sources and analytical approaches to assess the model's long-term efficacy in fostering durable cognitive development and career readiness.

Our initial longitudinal data analysis yielded promising insights into the long-term efficacy of the model. The alumni survey results (2018–2023 cohorts, $n=214$) showed that more than 80% of the respondents reported positive long-term effects on problem-solving skills, adaptability, and career advancement. The employment data analysis demonstrated that graduates of competency-based programs have higher employment rates and an increased likelihood of placement in leading technology companies.

Qualitative interviews with the alumni ($n=52$) revealed the persistent benefits of practical experience, collaborative learning skills, moral education, and social responsibility. Employer feedback ($n=30$) indicated that the graduates exhibited superior problem-solving abilities, teamwork, and innovative thinking skills. Longitudinal cognitive ability assessment ($N=100$) showed that compared to the control group using traditional teaching methods, graduates demonstrated higher cognitive flexibility and enhanced analytical skills that were sustained for up to three years after graduation ($p<0.01$), indicating a lasting impact on higher-order cognitive functions.

Based on these preliminary longitudinal findings, we developed several short-term research directions. We plan to conduct periodic skill assessments to measure how well graduates maintain and apply the programming and cognitive skills learned during their studies, tracking skill level changes within to 1–3 years of graduation. We are creating detailed career trajectory maps to analyze graduates' career choices and progression in the first to 3–5 years after graduation. Additionally, we initiated a 3-year longitudinal study tracking a cohort of graduates to assess the model's impact on cognitive abilities during the early career stage.

These preliminary longitudinal findings provide compelling evidence for the sustained efficacy of competency-based teaching models in fostering cognitive development, career readiness, and holistic professional growth. Ongoing longitudinal data analysis is expected to provide nuanced insights into how pedagogical approaches influence long-term professional success in the rapidly evolving computer science field.

5 Discussion and conclusion

5.1 Interpretation of key findings

Our findings provide compelling evidence for the efficacy of a competency-based teaching model in enhancing students' cognitive and professional abilities in undergraduate programming courses. The results demonstrated statistically significant improvements across multiple dimensions including problem-solving skills, CT, and SDL capabilities.

The empirical validation of our theoretical framework provides several key insights. Constructivist and situated learning theories have the strongest impact on practical skill development, whereas cooperative and SDL theories significantly enhance learning autonomy. Educational and humanistic theories have contributed to holistic development. The integrated framework demonstrated superior outcomes compared with single-theory approaches, with inter-theory interactions creating additional learning benefits and supporting comprehensive skill development.

5.2 Theoretical implications

The observed outcomes not only corroborate but significantly extend existing theoretical frameworks in educational psychology and cognitive science. Our findings demonstrate strong alignment with constructivist learning theory (Piaget, 1976), confirming that active knowledge construction through programming activities enhances cognitive schema development. Additionally, the results support cognitive load theory (Sweller, 1988), illustrating how structured competency-focused instruction effectively optimizes cognitive resource allocation by managing intrinsic complexity while reducing extraneous cognitive demands, thereby facilitating deeper conceptual understanding and skill development.

The significant improvements in cognitive abilities, as measured by R.SPM scores, support Cattell's (1963) Investment Theory, which posits that fluid intelligence contributes to the development of crystallized intelligence through the learning experience. The strong correlations between cognitive enhancement and domain-specific skills are consistent with Ericsson's (2006) theory of deliberate practice in expertise development.

The identified predictors and mediating factors offer valuable insight into the refinement of pedagogical strategies in computer science. The role of classroom engagement, practical project completion, and participation in innovative activities in predicting cognitive improvement aligns with Vygotsky's (1978) sociocultural theory of cognitive development, which emphasizes the importance of active contextualized learning experiences.

5.3 Practical implications

The practical implications of this study are multi-faceted. The model presents a paradigm shift in programming education, moving from a content-centric to a com-

petency-centric approach. Our findings underscore the importance of explicitly integrating problem-solving and CT skills into the curriculum structure.

Mediation analysis suggested that enhancements in general cognitive abilities facilitated the development of CT and problem-solving skills, which subsequently contributed to the overall learning performance. The interplay between various components of the model underscores the significance of a holistic approach to programming education. The model's efficacy lies in its ability to cultivate a range of interconnected competencies rather than focusing on isolated skills.

These findings extend the current understanding of effective programming education methodologies, demonstrating how a well-designed competency-based model can foster not only programming skills but also broader cognitive and professional abilities. This aligns with and builds upon previous research emphasizing the importance of integrated approaches in STEM education (e.g., Johnson et al., 2020; J. Al Quraan & A Forawi, 2019).

5.4 Adaptation scope, limitations, and future research directions

5.4.1 Adaptation scope

The model is primarily designed for foundational undergraduate computer programming courses, particularly core courses in the first and second years of study. It is most suitable for students with a certain level of SDL ability and motivation, especially those with a strong interest in programming. The educational environment requires well-equipped computer laboratories, a stable network infrastructure, comprehensive online learning platforms, and programming judge systems. Faculty members need strong professional capabilities and willingness to adopt innovative teaching methods. While developed for programming courses, the model may be adaptable to other practice-oriented STEM courses such as data structures and algorithm design.

5.4.2 Limitations of the model

While this investigation provides substantial empirical evidence supporting the effectiveness of our competency-based teaching model, several methodological limitations warrant acknowledgment. First, despite the substantial sample size, the single-institution context may constrain generalizability to institutions with different resources, faculty expertise, or student demographics. Second, although our quasi-experimental design incorporated multiple control measures, the inability to implement full randomization limits definitive causal inferences. Third, while the observed correlations between intervention components and outcomes are robust, potential unmeasured confounding variables cannot be completely eliminated. The quasi-experimental design, necessitated by real-world educational constraints, may have introduced potential bias.

Although preliminary longitudinal data have shown positive long-term effects, more extended follow-up studies are needed to confirm the sustainability of these effects, particularly regarding students' career development and lifelong learning

skills. More rigorous comparative studies are needed to fully establish the effectiveness of the model compared with other teaching methods.

This model may not be suitable for short-term intensive courses, and may pose challenges for beginners. Moreover, the high demand for teaching equipment and online platforms may limit their applicability in resource-constrained educational institutions. This model requires teachers to be adaptable and willing to apply new teaching methodologies, which may cause resistance among the faculty members.

5.4.3 Future research directions

To address these limitations and develop the model further, future research should focus on applying it to larger and more diverse educational settings, extending the study duration to assess long-term effects, and using reliable metrics to compare the competency-based model with other teaching methods. Researchers should explore the effectiveness of the model across different student groups and environmental factors, develop best-practice guidelines and policy recommendations for nationwide programming education reforms, and investigate its adaptability to different STEM disciplines beyond programming.

Creating simplified versions of the model for various educational environments and student groups, conducting long-term tracking studies, and exploring the integration of AI technologies into the model are important areas of future research. In addition, future research should focus on external validation through peer reviews and expert feedback, adaptability assessment across various learning environments, and establishing regular communication with employers to assess how graduates' competencies align with their industry needs.

5.5 Dynamic improvement strategies

Throughout the 12-year implementation period, we systematically employed data-driven, iterative improvement strategies within a continuous quality enhancement framework. This methodical approach ensured the model's responsive adaptation to evolving educational requirements, technological advancements, and emerging industry needs. Based on student feedback in 2018 indicating difficulty in understanding pointer concepts, we introduced additional visualization tools and interactive exercises in 2019, resulting in a significant improvement in students' understanding (a 15% increase in related test scores).

As the importance of AI in programming increased, we integrated basic machine learning concepts into the curriculum in 2021, introducing a small machine learning project using the C language. During the 2020 pandemic, we quickly adjusted the teaching mode to accommodate online learning, developing a series of short online coding challenges that maintained student engagement above 90% with a course completion rate of 95%.

Based on employer feedback in 2019 and the identification of deficiencies in graduates' large project management skills, we introduced a semester-long team project in 2020 to simulate a real software development cycle. In 2017, the analysis indicated that traditional written exams might not comprehensively assess students' practical

programming abilities, leading to the introduction of project-based assessments in 2018, accounting for 40% of the total grades. In 2022, we began using learning analytics to identify each student's strengths and weaknesses, developing personalized learning plans that increased average student grades by 8% within a year while significantly reducing grade disparities.

These dynamic improvement strategies not only enhanced teaching quality but also ensured that the model could continuously adapt to student needs and industry changes. By consistently collecting feedback, analyzing data, and implementing improvements, our teaching model maintains its relevance and effectiveness.

5.6 Conclusion

This 12-year longitudinal investigation establishes compelling empirical evidence for the substantial efficacy of an integrated competency-based teaching model in enhancing students' comprehensive abilities and cognitive development in university programming education. The documented improvements in cognitive measures, academic performance, and innovation capabilities present significant implications for transforming computer science education methodologies and contributing to sustainable technological workforce development in an increasingly digital society.

The cognitive skills cultivated through this model have profound implications for students' long-term career prospects and their potential societal contributions. Enhanced problem-solving abilities, critical thinking, and adaptability equip graduates to navigate a rapidly evolving technological landscape, fostering innovation in fields such as AI and cybersecurity. The ethical considerations integrated into our model prepare students to be responsible technology leaders, addressing the growing need for ethical reasoning in technological advancement (Tanguay, 2021).

By equipping students with essential cognitive and collaborative skills for the digital age, the model contributes to addressing the skill gap identified in industry reports (World Economic Forum, 2023) and prepares students for future employment landscapes. This study not only provides an empirical foundation for the reform of programming education but also offers valuable insights for broader STEM education innovation. By fostering students' cognitive abilities, problem-solving skills, and ethical awareness, our model contributes to the development of the next generation of professionals capable of addressing the technological challenges of the 21st century.

Author contributions DY and JL, were in the conceptualization of the study. DY was responsible for data collection. DY and JL performed data analysis. DY first initiated the drafting of the manuscript. JL proof-read and revised the manuscript. All the authors contributed to the study.

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Data availability The data used to obtain the findings of this study are available from the corresponding author upon request.

Declarations

Ethical approval Participation was voluntary and anonymous. Informed consent was obtained from all the participants.

Competing interests The author declares that she has no competing interests.

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Authors and Affiliations

Dunhong Yao^{1,2,3}  · Jing Lin^{1,2,3} 

✉ Jing Lin
linjing@hhtc.edu.cn

Dunhong Yao
dh_yao@hhtc.edu.cn

¹ School of Computer and Artificial Intelligence, Huaihua University, Huaihua 418000, China

² Key Laboratory of Intelligent Control Technology for Wuling-Mountain Ecological Agriculture in Hunan Province, Huaihua 418000, China

³ Key Laboratory of Wuling-Mountain Health Big Data Intelligent Processing and Application in Hunan Province Universities, Huaihua 418000, China